

Question 1:

The problem can be broken down into 4 parts:

1. Initializing A for Hough space (1 point)
 - Operator #4 in published version
2. Voting in Hough space for parameters which define squares (2 points)
 - Each black point should contribute votes to all the pixels in A which are at a **D8** distance of 2 from it. Squares are “circles” in D8, we therefore essentially vote for all possible centers at a distance of $r=2$ (op #20).
3. Choosing a threshold in A (2 points)
 - Since >75% of the pixels in each square are black, we’re looking for points in A with >12 votes (op #7).
4. Whitening pixels in original image based on the points left after threshold (2 points)
 - What’s left is to whiten points at D8 distance of 2 from the points above the threshold (op #11).

Those who had a “block” of the above four operations in succession which included at least one voting/whitening with D8 received an extra point, so a complete answer was worth 8 points.

Reductions:

The following mistakes reduced one point each:

- Voting using D4, lines or euclidean
- Wrong threshold
- Whitening using D4, lines or euclidean
- Closing (vertical)
- Closing (horizontal)
- Extra voting/whitening/other unnecessary operations

Question 2:

Each section is worth 2 points.

- A. Huffman (LZ needs sequences, which are most probably not present here).
- B. 4 bits (16 different numbers)
- C. Black (the color yellow absorbs blue, so there would be no light reflected back).
- D. 29×29 ($128 \rightarrow 126 \rightarrow 63 \rightarrow 61 \rightarrow 31 \rightarrow 29$: pooling on a dimension which is not divisible by it is rounded up, as we don’t want to throw away information).
- E. **We made a mistake in this question. The correct answer is 18×18 and did not appear as one of the possibilities. We counted 17×17 as the “correct” answer**

because during the test we said to mark the “closest” answer. Anyone who marked something else, but has ‘18x18’ written somewhere in the notebook will receive full point for this question.

- F. Quadrants $H_x L_y$ and $H_x H_y$ will be zero, because taking the derivative (high pass) in the x direction gives zero (pixel values along rows are constant). Partial score (1 point) was give to those with 3/4 correct marks.